

4.7.4. Use proof by contradiction to show that for every integer m , $7m + 4$ is not divisible by 7.

Given statement is that $\forall m \in \mathbb{Z}, 7 \nmid (7m + 4)$

Proof:

Suppose not. Suppose m is some integer.

We must show that $7m + 4$ is divisible by 7.

$$\exists m \in \mathbb{Z} \text{ st } 7 \mid 7m + 4$$

Definition of divisibility:

$$d \mid n \leftrightarrow \exists k \in \mathbb{Z} \text{ such that } n = dk$$

Let $m \in \mathbb{Z}$

$$d = 7$$

$$n = 7m + 4$$

$$7 \mid 7m + 4 \leftrightarrow 7m + 4 = 7k, \text{ for some } k \in \mathbb{Z}$$

$$7m = 7k - 4$$

$$4 = 7k - 7m = 7(k - m)$$

$$4/7 = k - m$$

Since $k - m \in \mathbb{Z}$, but $4/7 \notin \mathbb{Z}$, the statement is true by proof by contradiction, which is what was to be shown.

□

Solution: Therefore, $7m + 4$ is not divisible by 7.